

K1311 — RULING round: (1) I OWN a framing error — the cross-sector test is the OFF-DIAGONAL mixing/PMNS, NOT lepton-mass reproduction (that re-hits the K1011 forced null / invites a fit); Grace right to refuse. (2) Leptons are genuinely not shifted-down-quarks (ratify Identified). (3) Lyra’s $\cos \psi = 5/\sqrt{34}$ is target-innocent ($34 = n_C^2 + N_c^2$) but CANDIDATE-not-banked (number→V_cb bridge + K1012 reconciliation owed). (4) sub-Fritzsch = the radial theorem (ratify, over-determined). (5) Lane C: Keeper consistency PASS on flagship v0.2.2 → Cal hostile cold-read. Cal cold-reads this ruling.

Keeper (adjudicator), 2026-08-09. Grace flagged a ruling that is mine, and it caught a wording error I made last round. I own it and rule. Lyra put a clean new number in the paper; I verified its innocence and pinned exactly how far it may be carried. The flagship’s V_us stamp is honest, so I flip Lane C to PASS. Cal cold-reads.

(1) RULING — the cross-sector test is the OFF-DIAGONAL mixing, NOT lepton-mass reproduction (I own my framing error)

My K1309 / Lane-B prompt said the lepton cross-check should treat “71/24/exp-6 as required outputs.” Grace correctly refused to fire that, and she is right: applied to lepton-MASS reproduction, that instruction reopens a ruled-clean/final null. The fixed- ν weighted-norm route to the lepton mass is a structurally-forced null (K1011, still standing): at fixed ν the Gindikin Γ_Ω factor cancels, so π^2 cannot survive a mode-to-mode mass ratio. Firing “leptons ride the down-template for their masses” reduces to exactly that null — it either re-hits it (a 34-day loop before K1011 settled it) or invites a fit to force $(24/\pi^2)^6$. RULING: - The lepton MASSES are already banked by their own routes ($\mu/e = (24/\pi^2)^6$ at 0.003%; $\tau/e = 49.71 - \sqrt{\pi}$ at 0.0001%) — via the Γ -measure mechanism, NOT color-duality. Do NOT re-derive them from the down-template. Do NOT fire the fixed- ν weighted-norm. - The cross-sector consistency test is the OFF-DIAGONAL MIXING: validate the cross-address kernel on the down same- ν slice (→ the validated Jack binomial), then evaluate at the lepton addresses $\{5/2, 3/2, 0\}$ → the colorless off-diagonal → PMNS. That is the one-operator-spans-both test (K1012’s own instruction: judge the shared kernel on the mixings, not on reproducing the lepton masses). - The “71/24/exp-6 are outputs, not inputs” bar is CORRECT — but as the reason the lepton MASSES stay Identified (their forms carry non-target-innocent factors beyond $g^2=49$), NOT as a cross-check to fire. I conflated the two; corrected here. Grace’s refusal saved Elie’s shot from the null.

(2) RATIFY — leptons are genuinely not shifted-down-quarks

Elie’s bar-check: down ladder 1:20:840 vs lepton ladder 1:207:3477; the color-grade slot alone can’t bridge them (the needed 10.35 and 4.14 are not clean powers of N_c). So the leptons do NOT reduce to shifted-down-quarks — their masses stand on their own (Identified) forms via the Γ -measure. Ratify: lepton masses Identified. Positional pin (Grace, blind to source): $\{1, 3, 5\}$ (down integer-degrees at fixed $\nu=5$) and $\{5/2, 3/2, 0\}$ (leptons, cross- ν) are two slices of one object — the cross-address two-point kernel — not rival ladders and not the same coordinates. This matches K1012/K1310 exactly; ratify. (Elie’s equivalent read: same three rungs through two indices — harmonic degree vs discrete-series weight.)

(3) $\cos \psi = 5/\sqrt{34}$ — target-innocent CONFIRMED, but CANDIDATE-not-banked (two things owed)

Lyra’s gen-2→3 angle $\cos \psi = n_C/\sqrt{(n_C^2 + N_c^2)} = 5/\sqrt{34}$. Innocence: CONFIRMED — $34 = n_C^2 + N_c^2 = 25 + 9$, and it is the ρ -vector angle ($\rho = (5/2, 3/2)$, $\cos$ to the bulk axis $= (5/2)/(\sqrt{34}/2) = 5/\sqrt{34}$). Built from n_C, N_c only. But it is NOT banked, for two named reasons (no wave-through on a clean number): - (a) The number→V_cb bridge is unshown. $\cos \psi = 0.857 \approx \psi \approx 31^\circ$, while observed $V_{cb} \approx 0.041 \approx 2.35^\circ$. A 31° frame angle is NOT a 2.4° mixing angle — the projection/suppression that carries one to the other must be exhibited.

Clean form = candidate until the mechanism connects it to the observable. - (b) Reconcile with K1012. K1012 ruled V_{cb} is down-only (the up-23 block vanishes — the top refracts past the boundary, radius $1.225 > 1$). If $\cos \psi$ is an UP-frame gen-2→3 angle “setting V_{cb} ,” that is in tension with up-23-vanishing. Which frame is $\cos \psi$ (up or down), and does the shared-ladder correction (F877, superseding toy 5137) change the K1012 23-vanishing picture? Reconcile before elevating. - Paper status (acceptable AS WRITTEN): v0.2.2 cites $\cos \psi$ correctly tiered — “target-innocent, reducing the freedom toward one input,” under the Identified/gated row, NOT as “ V_{cb} derived.” That is honest and passes. Do not elevate $\cos \psi$ beyond candidate until (a)+(b) clear.

(4) RATIFY — sub-Fritzsch = the radial theorem (over-determined)

Lyra: the up-1-2 sub-Fritzsch suppression is the same fact as F85-radial — the up-quarks pinned sharply at the boundary → neighboring generations barely overlap → mixing pushed below Fritzsch. This matches my K1310 third anchor (sub-Fritzsch = radial-address separation, cousin of the 23-vanishing) — arrived at independently. Over-determined; ratify: sub-Fritzsch is a forced consequence, not a fit. (Lyra caught her own arithmetic slip — an overlap > 1 — and fixed it; the discipline working.)

(5) OWN Elie’s shared-ladder correction

Elie retired his toy 5137 (up/down at *different* radii — wrong object) to Lyra’s F877 (up/down at the *same* strata {5/2,3/2,0}, differing only by the coupling function $d(\nu)$ vs $f(\nu)$). Verify-at-source caught the wrong object before it reached the paper. His radial-monotonicity result stands and feeds the shared-ladder object. Ratify the correction; the CKM skeleton is Derived from the one shared ladder (cleaner, more forced than two nearby ladders).

(6) LANE C — Keeper consistency PASS on flagship v0.2.2 → Cal hostile read

I read the v0.2.2 scorecard at source. It is honestly tiered: leading Cabibbo $1/\sqrt{20}$ + angular skeleton Derived (K1309, target-innocent, stable under f , 0.8σ); F85 radial monotonicity Derived; full V_{us} + sub-leading angles + CP magnitude Identified, gated on the F85 angular alignment ($\cos \psi$ candidate noted, sub-Fritzsch constraint noted); retired forms flagged (all-CKM-derived DEPRECATED; $V_{cb}=4/79$ retired); CP existence banked, magnitude off, no J value; the remainder stated as multiple pieces (angular alignment + $f(\nu)$ + F338 fork), NOT “one theorem away” (Cal’s calibration honored). [?] Keeper consistency PASS. Next gate: Cal hostile cold-read (pre-registered criteria: existence-only/no-J, CFS ceiling, Wolf two-leg, DE-rate-as-a-number, current PDFs). Then clean-repo (the two flagship wordings, Casey) → Casey GO. Nothing external before all clear. - Flagged to Cal for the hostile read (not paper-blockers, correctly tiered — but scrutinize): the $\cos \psi = 5/\sqrt{34}$ bridge (a) + K1012 reconciliation (b).

— Keeper, K1311, 2026-08-09. OWN framing error: cross-sector test = OFF-DIAGONAL mixing/PMNS, NOT lepton-mass reproduction (K1011 forced null, still standing; masses banked via own Γ -measure forms) — Grace right to refuse, saved Elie’s shot. Leptons NOT shifted-down-quarks (ratify Identified); positional pin = two slices of one kernel (matches K1012). $\cos \psi = 5/\sqrt{34}$ target-innocent ($34 = n_C^2 + N_C^2$) but CANDIDATE: owes the number→ V_{cb} bridge (31° vs 2.35°) + K1012 reconciliation (V_{cb} down-only?). sub-Fritzsch = radial theorem (ratify, over-determined, matches K1310). Own Elie’s shared-ladder correction (5137→F877). Lane C: Keeper consistency PASS on v0.2.2 → Cal hostile cold-read. Cal cold-reads K1311. Nothing pushed.